1 Algebra and Equations

The big idea. Algebra is *the language of unknown numbers*. Instead of saying at length "some number, add 5, then double it", we simply write $2(x+5)$. The letter x (called a **variable** or an **unknown**) holds the place of a number that we do not know yet or that may change.
In this topic you will practise four core skills:

1. **Translate** words into expressions and formulas ("more than", "times", "sum", "difference"...).
2. **Substitute** numbers to evaluate an expression, or to check whether an equation is true or false.
3. **Solve equations** with the *balance principle*: whatever you do to one side, do exactly the same to the other side.
4. **Spot patterns**: sequences, tables of values, ratios, and the order of numbers.

Every word problem follows the same route: *read carefully* → *choose an unknown* → *write an expression or an equation* → *solve* → *check* → *answer*.

① Variables, algebraic expressions and formulas

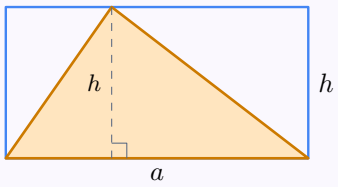
1. Variables and expressions. An **algebraic expression** is built from numbers, letters (variables) and the operations $+$, $-$, $\times$, $\div$, powers and brackets. Examples: $3x-1$; $2(n-1)$; $120+30m$; xy.
Short-hand. Between a number and a letter, or between two letters, we leave out the multiplication sign: $3x$ means $3\cdot x$; xy means $x\cdot y$; $2(n-1)$ means $2\cdot(n-1)$. The number is written before the letter.
2. Value of an expression. To evaluate an expression for given values of the variables, we *replace* each letter by its number (best put in brackets), then calculate in the correct order of operations.
Example: for $x=3$, $3x-1=3\cdot 3-1=9-1=8$; $2x^2=2\cdot 3^2=2\cdot 9=18$.
3. Translating words into expressions.

Words	Expression
5 more than x; x increased by 5	$x+5$
5 less than x; x decreased by 5	$x-5$
3 times x; triple x	$3x$
half of x; x divided by 2	$\frac{x}{2}$ or $x\div 2$
start with 120, add 30 each time, m times	$120+30m$
three consecutive whole numbers, the smallest is n	$n,\ n+1,\ n+2$

4. A formula is an equation that tells us how to calculate one quantity from other quantities. Useful geometry formulas (side lengths a, b, c):

- Perimeter of a triangle: $P = a + b + c$ (the sum of the three sides).
- Rectangle with length a and width b: perimeter $P = 2(a + b)$, area $A = a \cdot b$.
- Triangle with base a and height h: area $A = \dfrac{a \cdot h}{2}$.
- Rectangular prism $a \times b \times c$: total surface area $S = 2(ab + bc + ca)$, volume $V = abc$.

Why is it true? Why is the area of a triangle half of (base × height)? Draw a rectangle with one side equal to the base a of the triangle and the other side equal to the height h. The height splits the triangle into two right triangles; each right triangle is exactly half of a smaller rectangle (in the picture, the shaded and white parts match in pairs). So the triangle covers exactly half of the big rectangle: $A = \dfrac{a \cdot h}{2}$.



Similarly, the perimeter of a rectangle is $a + b + a + b = 2a + 2b = 2(a + b)$ because a rectangle has two equal long sides and two equal short sides. □

ThinkAlgo Tip. When substituting, *always put the number in brackets*, especially a negative number: for $x = -2$, $x^2 = (-2)^2 = 4$, while $3x = 3 \cdot (-2) = -6$. Brackets protect you from sign mistakes.

Common mistake. $3x$ with $x = 4$ is $3 \cdot 4 = 12$, *not* "34". And x^2 (which is $x \cdot x$) is different from $2x$ (which is $x + x$): for $x = 5$, $x^2 = 25$ but $2x = 10$.

ThinkAlgo Tip. Write everything in terms of a few letters. Consider a list in which every number (from the third on) is the sum of the two before it. If the first two numbers are a and b, the next ones are

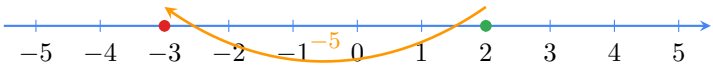
$$a, \quad b, \quad a + b, \quad b + (a + b) = a + 2b, \quad (a + b) + (a + 2b) = 2a + 3b, \ \ldots$$

For instance, with $a = 11$, $b = 20$ the fifth number is $2 \cdot 11 + 3 \cdot 20 = 22 + 60 = 82$, just as in the list $11, 20, 31, 51, 82$. Writing the numbers in terms of a and b turns a question about the list into an equation.

Note for parents. Play the "number machine" game with your child: you say a rule ("multiply by 3, then subtract 1"), your child writes it as the expression $3x - 1$, and then you take turns feeding numbers into the machine. This game makes the letter x feel friendly instead of scary.

② Integers, order of operations and the distributive law

1. Integers. The numbers $\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$ are called integers. On the number line, a number further to the left is smaller. The opposite of a is $-a$; we have $a + (-a) = 0$.



Example: $2 + (-5) = -3$: starting at 2, move 5 steps to the left and arrive at -3.

2. Adding and subtracting integers.

- Two numbers with the same sign: add the sizes and keep the common sign: $(-11) + (-26) = -37$.
- Two numbers with different signs: subtract the smaller size from the larger one and take the sign of the number with the larger size: $11 + (-37) = -26$.
- Subtracting means adding the opposite: $a - b = a + (-b)$. Example: $-26 - 11 = -26 + (-11) = -37$.

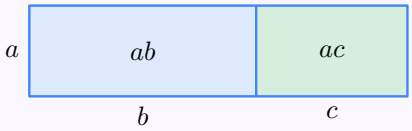
3. Order of operations. Brackets first → powers → multiplication and division (left to right) → addition and subtraction (left to right). Example: $6 + 2 \cdot 3 = 6 + 6 = 12$ (not $8 \cdot 3$).

4. The distributive law. $a(b+c) = ab+ac$ and $a(b-c) = ab-ac$. *Examples:* $3(T+4) = 3T+12$; $20 \cdot 9\,000 - 20 \cdot 6\,750 = 20 \cdot (9\,000 - 6\,750)$.

5. Simplifying expressions. Add "like terms": $3x+5x = 8x$; $(a+1)+(b+1)+(c+1)+(d+1) = (a+b+c+d)+4$.

6. Points in the coordinate plane. The point (x, y) has x-coordinate x and y-coordinate y. Moving k units to the right increases x by k; moving k units up increases y by k (moving left or down decreases them).

Why is it true? Why is $a(b+c) = ab+ac$? Take a rectangle with width a and length $b+c$. Its area is $a(b+c)$. But we can also cut it into two smaller rectangles with areas ab and ac. It is the same area, so $a(b+c) = ab+ac$.



□

Common mistake. $3(T+4)$ is *not* equal to $3T+4$: you must multiply 3 by *each* term inside the brackets, $3(T+4) = 3T+3 \cdot 4 = 3T+12$. Likewise $-(a-b) = -a+b$ (change the sign of *every* term in the brackets).

③ Equations and the balance principle

1. Equations. A statement of equality containing an unknown, such as $2(x+8) = 38$, is called an **equation**. A number that makes both sides equal when it replaces the unknown is called a **solution**. To *solve an equation* means to find its solutions.

2. The balance principle. Imagine the equation as a pair of scales in balance. The scales stay balanced if we:

- add (or subtract) *the same number* on both sides;
- multiply (or divide) both sides by *the same non-zero number*.

A familiar consequence is the **rule for moving terms**: when a term moves from one side to the other, its *sign changes*.



Example: two identical boxes plus a 3 kg weight balance 11 kg: $2x + 3 = 11$. Remove the 3 kg weight from both pans: $2x = 11 - 3 = 8$. Halve both pans: $x = 8 \div 2 = 4$. Each box weighs 4 kg.

3. Substitution on scales. If A on one pan balances B, then on any other pair of scales we may replace A by B and the scales stay balanced (just like replacing an expression by an equal expression).

4. Always check. After solving, substitute the solution into the original equation to make sure both sides are equal.

Why is it true? Why does a term change sign when it moves across? From $2x+3 = 11$, subtract 3 from both sides (the scales stay balanced): $2x + 3 - 3 = 11 - 3$, that is, $2x = 11 - 3$. Looking at the result, the term $+3$ on the left seems to have "jumped" to the right and become -3. In fact nothing "jumps": we only subtracted the same number from both sides. □

Common mistake. Forgetting to change the sign: from $x + 11 = -26$, writing $x = -26 + 11$ is **wrong**. The correct step is $x = -26 - 11 = -37$.

ThinkAlgo Tip. In a multiple-choice question with an equation, you can *substitute each option in turn* into the equation. It is slower but very safe, and it is also the best way to check.

④ Solving word problems with equations

Four steps for a word problem.

1. **Choose an unknown**: call the quantity you want (or a related one) x; write its unit and any condition (for example $x > 0$, x is a whole number).
2. **Express** the other quantities in terms of x.
3. **Write an equation** from the conditions of the problem and **solve** it.
4. **Compare** with the conditions, **check**, and **answer** the actual question.

Common types.

- **Consecutive integers:** $n,\ n+1,\ n+2$ (or, more neatly, $n-1,\ n,\ n+1$). Consecutive even (odd) numbers differ by 2: $n,\ n+2,\ n+4$.
- **Sums in pairs:** three objects a, b, c weighed two at a time give $a+b$, $b+c$, $c+a$. Adding the three results: $(a+b)+(b+c)+(c+a) = 2(a+b+c)$, so *the total of the three objects is half*

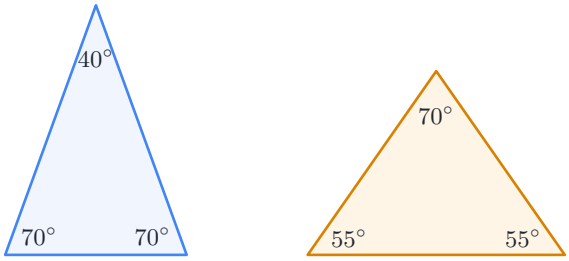
of the sum of the results. To find one object, subtract the pair formed by the other two from the total.

- **Angles in a triangle:** the three angles of a triangle add up to 180°; an isosceles triangle has two equal base angles.
- **Average:** average = sum of the values ÷ number of values; average price = total cost ÷ total mass.

Two small examples.

a) An isosceles triangle with a 70° angle. The 70° angle may be a *base angle* or the *apex angle*, so we must look at two cases:

- If 70° is a base angle: the base angles are 70° and 70°, and the apex angle is $180° - 70° - 70° = 40°$.
- If 70° is the apex angle: the two base angles are equal and add up to $180° - 70° = 110°$, so each is 55°.



b) Average price. You buy 300 g of type I candy for 15 000 dong and 200 g of type II candy for 20 000 dong. The average price of 100 g of candy is $(15\,000+20\,000)\div(300+200)\cdot 100 = 35\,000\div 500\cdot 100 = 7\,000$ dong. You must *not* average the two prices 5 000 and 10 000 dong per 100 g (that gives 7 500 dong, which is wrong), because different amounts of the two candies were bought.

Why is it true? Three consecutive integers: the middle one equals the sum divided by 3. Call the three numbers $n-1$, n, $n+1$. Their sum is $(n-1)+n+(n+1) = 3n$ (the -1 and $+1$ cancel). So the middle number is n = sum ÷ 3. For example, if the sum is 153, the middle number is 51 and the three numbers are $50, 51, 52$. □

ThinkAlgo Tip. Choosing the *middle number* as the unknown usually gives the neatest equation. For "sums in pairs" problems, do not rush to use three unknowns: *add everything up* first.

Common mistake. Solving the equation is not the end of the problem! Re-read the question: does it ask for the *largest* number, the *lightest* object, or the *total*...? Many students find the correct x but choose the wrong option because they answer a different question.

Note for parents. When your child is stuck, ask: "What don't we know yet? Let's call it x. Now, what does the problem tell us about x?" Let your child say it in words first, and only then write it as an equation.

⑤ Patterns, tables of values and arithmetic sequences

1. Arithmetic sequences. A list of numbers in which neighbouring numbers always differ by the same number d (called the **common difference**) is called an arithmetic sequence. If the first term is a, then

$$\text{the } n\text{-th term } = a + (n-1) \cdot d.$$

Example: the sequence $7, 10, 13, 16, \ldots$ has $a = 7$, $d = 3$; its n-th term is $7 + 3(n-1) = 3n + 4$.

2. From a table of values to an expression. If each time n increases by 1 the value increases by d, the expression has the form $d \cdot n +$ (some number). Find "some number" by substituting $n = 1$.

n	1	2	3	4	5
value	4	7	10	13	16

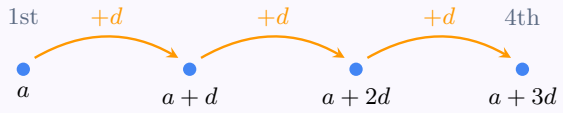
Each time n increases by 1, the value increases by 3, so the expression has the form $3n + b$. Substituting $n = 1$: $3 + b = 4$, so $b = 1$. Hence value $= 3n + 1$.

3. Is a number in the sequence? The number N belongs to the sequence $a,\ a+d,\ a+2d, \ldots$ exactly when $N-a$ is divisible by d (and $N \geq a$). For a sequence counting down, $a,\ a-d,\ a-2d, \ldots$, look at $a - N$ instead.

4. Common terms of two sequences; repeating events. If two arithmetic sequences with differences d_1 and d_2 have a common term, then their common terms are exactly $\text{lcm}(d_1, d_2)$ apart. Similarly, events that repeat every p minutes and every q minutes happen together again every $\text{lcm}(p, q)$ minutes.

ThinkAlgo Tip. *An example of repetition.* The first bus returns to the station every 12 minutes, the second bus every 20 minutes. At 7:00 a.m. both are at the station. The first bus returns after $12, 24, 36, 48, \mathbf{60}, \ldots$ minutes; the second after $20, 40, \mathbf{60}, \ldots$ minutes. The two buses meet at the station again every $\text{lcm}(12, 20) = 60$ minutes, that is, at 8:00 a.m., 9:00 a.m., ... Similarly, if the two sequences $7, 10, 13, \ldots$ (difference 3) and $2011, 2006, 2001, \ldots$ (difference 5) have a common term, then their common terms are $\text{lcm}(3, 5) = 15$ apart.

Why is it true? Why is the n-th term $a + (n-1)d$? Going from the 1st term to the n-th term, we cross $n - 1$ "gaps", and each gap adds d. It is like walking from the 1st lamp post to the 5th: you only pass 4 gaps.



From the 1st term to the 4th term there are 3 steps of $+d$, so the 4th term is $a + 3d$. □

Common mistake. Checking a formula against only *one* column of the table is not enough. For example, for the table $1, 3, 5, 7, 9$, both $3n - 2$ and $2n - 1$ give the value 1 when $n = 1$, but for $n = 2$ we get $3n - 2 = 4 \neq 3$. Test *at least two or three columns.*

⑥ Ratios, units of measurement and uniform motion

1. Directly proportional quantities. If one scoop of food feeds 8 fish, then 2 scoops feed 16 fish and 4 scoops feed 32 fish: when the number of scoops is multiplied by some number, the number of fish is multiplied by the same number. We say y is **directly proportional** to x with constant k if $y = k \cdot x$. The usual method is the **unit method** (work out the value for 1 part first, then multiply).

2. Uniform motion. With s the distance, v the speed and t the time:

$$s = v \cdot t, \qquad t = \frac{s}{v}, \qquad v = \frac{s}{t}.$$

Use matching units: if v is in km/h, then s is in km and t is in hours.

Two people moving *for the same length of time*: whoever is k times as fast covers k times the distance.

Running in opposite directions around a circle: from one meeting to the next, the *total* distance run by the two people is exactly one lap. If the first person is k times as fast as the second, then during each such stretch the first runs $\frac{k}{k+1}$ of a lap and the second runs $\frac{1}{k+1}$ of a lap.

3. Converting units of time. 1 minute = 60 seconds; 1 hour = 60 minutes = $60 \times 60 = 3\,600$ seconds; 1 day = 24 hours; 1 week = 7 days. Therefore 1 week = 7×24 hours = $7 \times 24 \times 60$ minutes = $7 \times 24 \times 60 \times 60$ seconds.

4. Money. Number of notes = total amount ÷ value of one note. Difference after n times = $n \times$ (higher price − lower price).

ThinkAlgo Tip. Read a product such as $60 \times 60 \times 24 \times 7$ as a *chain of unit conversions*, from right to left: 7 (days in a week) ×24 (hours in a day) ×60 (minutes in an hour) ×60 (seconds in a minute). The result is the number of seconds in one week.

Why is it true? Why can we work out the difference "per time" and then multiply? By the distributive law: $20 \times 9\,000 - 20 \times 6\,750 = 20 \times (9\,000 - 6\,750) = 20 \times 2\,250$. Both ways always give the same result, but the second is much quicker to do in your head. □

Common mistake. In motion problems, read carefully "how much has been travelled" and "how much is left". After half of a 510 km trip, the distance left is $510 \div 2 = 255$ km, not 510 km.

⑦ Equivalent fractions and a fraction of a quantity

1. The basic property. Multiplying (or dividing) the numerator and the denominator of a fraction by the same non-zero number gives an equal fraction: $\frac{3}{7} = \frac{3 \cdot 9}{7 \cdot 9} = \frac{27}{63}$.

2. Recognising equal fractions. For $b \neq 0$, $d \neq 0$: $\frac{a}{b} = \frac{c}{d}$ exactly when $a \cdot d = b \cdot c$ ("cross-multiplying").

3. Fractions with the same numerator. If $\frac{1}{p} = \frac{1}{q}$ (or $\frac{m}{p} = \frac{m}{q}$ with $m \neq 0$), then $p = q$. Two equal fractions with equal numerators also have equal denominators.

4. A fraction of a quantity. To find $\frac{m}{n}$ of a number A, calculate $A \cdot \frac{m}{n} = A \div n \cdot m$. *Example:* $\frac{3}{7}$ of 168 km is $168 \div 7 \cdot 3 = 24 \cdot 3 = 72$ km. The rest corresponds to $1 - \frac{3}{7} = \frac{4}{7}$.

Why is it true? Why does $\frac{1}{p} = \frac{1}{q}$ imply $p = q$? Cross-multiply: $1 \cdot q = p \cdot 1$, that is, $q = p$. In other words: if a cake is cut into p pieces and another identical cake into q pieces, and the pieces are the same size, then the numbers of pieces must be the same. □

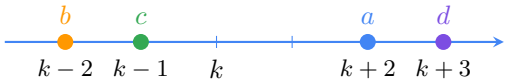
⑧ Comparing and reasoning about order

1. The transitive property. If $a < b$ and $b < c$, then $a < c$. This lets us line up several numbers: $a < b < c < d$.

2. The "set it equal to k" technique. When several expressions are equal, for example $a - 2 = b + 2 = c + 1 = d - 3$, call the common value k:

$$a = k + 2, \qquad b = k - 2, \qquad c = k - 1, \qquad d = k + 3.$$

Now every number is compared with the same reference point k; ordering the "extras" $-2 < -1 < +2 < +3$ we get $b < c < a < d$.



3. Logical reasoning about order. For sentences like "An owns more cats than Bình", translate them into symbols ($A > B$), write down all the conditions and link them with the transitive property. A statement that "*must be true*" has to be true in *every* situation that fits the problem; a single counter-example is enough to reject it.

ThinkAlgo Tip. *A reasoning example.* An is taller than Bình; Chi is shorter than Bình; Dũng is taller than Chi. Who is the shortest? In symbols: $A > B$, $B > C$, $D > C$. From $A > B > C$ and $D > C$ we see that A, B and D are all taller than C: **Chi is the shortest**. But we do *not* know whether Dũng is taller or shorter than An (for instance, if An is 1.6 m, Bình 1.5 m and Chi 1.2 m, then Dũng being 1.7 m or 1.3 m both fit all the conditions), so the statement "Dũng is the tallest" is *not necessarily* true.

Common mistake. Seeing that $a - 2$ and $b + 2$ are equal, many students conclude too quickly that "a is smaller because something is subtracted". The opposite is true: a must *lose* 2 to equal b *plus* 2, so a is exactly 4 more than b.

Note for parents. Use real objects: "If your bag loses 2 marbles, it has the same number as my bag plus 2 marbles. Which bag has more?" Let your child check with marbles. A concrete picture keeps the child from being fooled by plus and minus signs.

⑨ Worked examples

Example 1 (Translating words and evaluating an expression)

a) Evaluate the expression $A = 2x^2 - 3x + 1$ for $x = -2$.
b) Hoa has 50 000 dong in her piggy bank. Every week she puts in another 20 000 dong. Write an expression for the amount in the piggy bank after t weeks, then work out that amount after 6 weeks.

Guided solution.

Step 1. Part a: substitute $x = -2$, remembering the brackets

$$A = 2 \cdot (-2)^2 - 3 \cdot (-2) + 1.$$

Step 2. Calculate in order: powers → multiplication → addition and subtraction
$(-2)^2 = (-2) \cdot (-2) = 4$, so $2 \cdot 4 = 8$. $3 \cdot (-2) = -6$, so $-3 \cdot (-2) = +6$. Hence $A = 8 + 6 + 1 = 15$.

Step 3. Part b: find the pattern with a small table

Weeks	0	1	2	t
Amount (dong)	50 000	50 000 + 20 000	50 000 + 2 · 20 000	50 000 + 20 000t

The starting amount stays the same; the added part is 20 000 times the number of weeks. Expression: $50\,000 + 20\,000t$ (dong).

Step 4. Substitute $t = 6$
$50\,000 + 20\,000 \cdot 6 = 50\,000 + 120\,000 = 170\,000$ (dong).

⚠ **Common mistake.** Do not write $(50\,000 + 20\,000) \cdot t$: the starting amount is there only *once*; it must not be multiplied by the number of weeks.

Answer: a) $A = 15$; b) $50\,000 + 20\,000t$ dong; after 6 weeks there are 170 000 dong.

Example 2 (Solving equations with the balance principle)

Solve the equations: **a)** $5x - 7 = 2x + 11$; **b)** $\dfrac{x}{12} = \dfrac{5}{4}$.

Guided solution.

Step 1. Part a: collect the terms with x on the left
Subtract $2x$ from both sides: $5x - 2x - 7 = 11$, that is, $3x - 7 = 11$.

Step 2. Collect the numbers on the right
Add 7 to both sides: $3x = 11 + 7 = 18$.

Step 3. Divide both sides by 3
$x = 18 \div 3 = 6$.

✅ **Check:** Left side: $5 \cdot 6 - 7 = 30 - 7 = 23$. Right side: $2 \cdot 6 + 11 = 12 + 11 = 23$. Both sides are equal.

Step 4. Part b: use a common denominator (or cross-multiply)
We have $\dfrac{5}{4} = \dfrac{5 \cdot 3}{4 \cdot 3} = \dfrac{15}{12}$. The fractions $\dfrac{x}{12}$ and $\dfrac{15}{12}$ have the same denominator and are equal, so $x = 15$. (Cross-multiplying also works: $4x = 12 \cdot 5 = 60$, $x = 60 \div 4 = 15$.)

Recall. $\frac{a}{b} = \frac{c}{d}$ exactly when $ad = bc$ $(b, d \neq 0)$.

Answer: a) $x = 6$; b) $x = 15$.

Example 3 (Consecutive even numbers)

The sum of four consecutive even numbers is 116. Find the largest of the four numbers.

Guided solution.

Step 1. Choose an unknown

Let the smallest even number be n (n is even). Consecutive even numbers differ by 2, so the four numbers are $n,\ n+2,\ n+4,\ n+6$.

Step 2. Write an equation

$n + (n+2) + (n+4) + (n+6) = 116$, that is, $4n + 12 = 116$.

Step 3. Solve

$4n = 116 - 12 = 104$, $n = 104 \div 4 = 26$. The four numbers are 26, 28, 30, 32.

Check: $26 + 28 + 30 + 32 = 116$. Correct.

Step 4. Answer the actual question

The question asks for the *largest* number: it is 32.

ThinkAlgo Tip. Mental shortcut: the four numbers are equally spaced, so their average $116 \div 4 = 29$ lies exactly halfway between the two middle numbers 28 and 30. The four numbers follow at once.

Answer: 32

Example 4 (Weighing in pairs)

Three friends, Mai, Lan and Hoa, step on a scale two at a time. Mai and Lan weigh 72 kg; Mai and Hoa weigh 78 kg; Lan and Hoa weigh 80 kg. How many kilograms does each girl weigh?

Guided solution.

Step 1. Add all three weighings

In the three weighings, each girl is weighed exactly *twice*: $(M + L) + (M + H) + (L + H) = 72 + 78 + 80 = 230$, that is, $2(M + L + H) = 230$.

Step 2. Total mass of the three girls

$M + L + H = 230 \div 2 = 115$ (kg).

Step 3. Find each girl: subtract the pair that does not contain her from the total

Hoa $= 115 - (M + L) = 115 - 72 = 43$ kg; Lan $= 115 - (M + H) = 115 - 78 = 37$ kg; Mai $= 115 - (L + H) = 115 - 80 = 35$ kg.

Mai + Lan = 72 kg

Mai + Hoa = 78 kg

Lan + Hoa = 80 kg

Check: $35 + 37 = 72$, $35 + 43 = 78$, $37 + 43 = 80$. All three weighings match.

Answer: Mai 35 kg, Lan 37 kg, Hoa 43 kg.

Example 5 (Arithmetic sequences)

Consider the sequence 4, 7, 10, 13, … (each number is 3 more than the one before). **a)** Write an expression for the n-th term. **b)** Find the 50th term. **c)** Is 2026 in the sequence? If so, which term is it?

Guided solution.

Step 1. The n-th term

First term $a = 4$, common difference $d = 3$: the n-th term is $4 + 3(n-1) = 4 + 3n - 3 = 3n + 1$.

> **Recall.** The n-th term of an arithmetic sequence $= a + (n-1)d$.

Step 2. The 50th term

$3 \cdot 50 + 1 = 151$.

Step 3. Test the number 2026

We need a whole number n with $3n + 1 = 2026$, that is, $3n = 2025$. Since $2025 \div 3 = 675$ exactly, $n = 675$.

Check: $3 \cdot 675 + 1 = 2025 + 1 = 2026$. Correct.

> **Common mistake.** Do not forget to check that the division is exact. For instance, 2027 is *not* in the sequence because 2026 is not divisible by 3.

Answer: a) $3n + 1$; b) 151; c) yes, it is the 675th term.

Example 6 (Motion, fractions and unit conversion)

A train travels at a constant speed of 72 km/h on a 360 km route. It has already covered $\frac{2}{3}$ of the route. How much longer (in hours and minutes) does the train need to reach its destination?

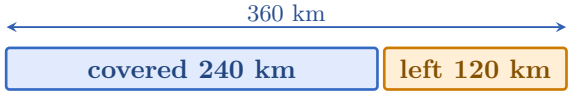
Guided solution.

Step 1. Distance already covered

$\frac{2}{3}$ of 360 km is $360 \div 3 \cdot 2 = 120 \cdot 2 = 240$ km.

Step 2. Distance left

$360 - 240 = 120$ km. (Or: the remaining part is $1 - \frac{2}{3} = \frac{1}{3}$ of the route, $360 \div 3 = 120$ km.)



Step 3. Time left

> **Recall.** $t = \frac{s}{v}$ (time = distance ÷ speed).

$t = 120 \div 72 = \dfrac{120}{72} = \dfrac{5}{3}$ hours $= 1\dfrac{2}{3}$ hours.

Step 4. Convert to hours and minutes

$\dfrac{2}{3}$ hour $= \dfrac{2}{3} \cdot 60 = 40$ minutes. So $t = 1$ hour 40 minutes.

Check: In 1 hour 40 minutes $= \dfrac{5}{3}$ hours, the train covers $72 \cdot \dfrac{5}{3} = 24 \cdot 5 = 120$ km, exactly the distance left.

Answer: 1 hour 40 minutes.

Example 7 (A balance-scale problem)

Both scales below are balanced. Identical shapes have equal weights.



How many circles ● balance two squares ■■?

Guided solution.

Step 1. Write each scale as an equation

Let S, T, C be the weights of a square, a triangle and a circle. Scale 1: $S = T + C$. Scale 2: $S + C = 3T$.

Step 2. Substitute scale 1 into scale 2

On scale 2, replace the square by "one triangle and one circle" (allowed by scale 1): $T + C + C = 3T$.

Step 3. Remove identical things from both pans

Remove one triangle from each pan: $C + C = T + T$, that is, $2C = 2T$, so $C = T$: a circle weighs the same as a triangle.

Step 4. Answer

$S = T + C = C + C = 2C$, so $2S = 4C$: two squares balance 4 circles.

Check: Try $C = T = 1$, $S = 2$: scale 1: $2 = 1 + 1$; scale 2: $2 + 1 = 3$. Both are correct.

Answer: 4 circles.

Example 8 (Ordering with the "set it equal to k" technique)

The positive integers a, b, c (all greater than 5) satisfy $\dfrac{2}{a+3} = \dfrac{2}{b-1} = \dfrac{2}{c+5}$. Arrange a, b, c in increasing order.

Guided solution.

Step 1. Equal fractions with the same numerator have equal denominators

The three fractions all have numerator 2 and are equal, so their denominators are equal: $a + 3 = b - 1 = c + 5$.

> **Recall.** If $\dfrac{m}{p} = \dfrac{m}{q}$ with $m \neq 0$, then $p = q$.

Step 2. Call the common value k

$a + 3 = k \Rightarrow a = k - 3$; $\quad b - 1 = k \Rightarrow b = k + 1$; $\quad c + 5 = k \Rightarrow c = k - 5$.

Step 3. Compare with the reference point k

$k-5 < k-3 < k+1$, so $c < a < b$.

Check: Take $k = 12$: $a = 9$, $b = 13$, $c = 7$. Then $\frac{2}{12} = \frac{2}{12} = \frac{2}{12}$ and indeed $7 < 9 < 13$.

Answer: $c < a < b$.

Core practice

Problem 1

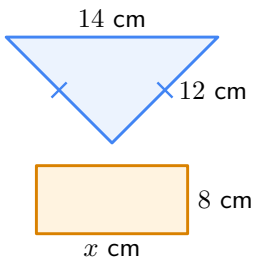
Mr Tâm drives a car at a constant speed of 85 km/h on a 510 km trip. He has now covered exactly half of the distance. How many more hours does he need to finish the trip?

(A) 5 hours (B) 425 hours (C) 12 hours (D) 1.5 hours (E) 3 hours

Your work:

Problem 2

In the diagram, the triangle has one side of length 14 cm and two equal sides of length 12 cm each; the rectangle has width 8 cm and length x cm. The perimeter of the triangle equals the perimeter of the rectangle. Find x.



(A) 8 (B) 10 (C) 11 (D) 14 (E) 15

Your work:

Problem 3

Let $x = 3$. Which of the following equations is true?

(A) $2x = 5$ (B) $3x - 1 = 8$ (C) $x + 5 = 3$ (D) $7 - x = 2$ (E) $6 + 2x = 14$

Your work:

Problem 4

The sum of three consecutive integers is 153. The largest of the three integers is

(A) 52 (B) 50 (C) 53 (D) 54 (E) 51

Your work:

Problem 5

One scoop of fish food can feed 8 goldfish. How many goldfish can 4 such scoops of fish food feed?

(A) 12 (B) 16 (C) 8 (D) 64 (E) 32

Your work:

Problem 6

If $x = 4$ and $y = 3x$, the value of y is

(A) 12 (B) 24 (C) 7 (D) 81 (E) 4

Your work:

Problem 7

Each bus ride costs Lan 6 750 dong, while Minh pays 9 000 dong per ride (they use different routes). If each of them rides the bus 20 times, how much less does Lan pay than Minh in total?

(A) 75 000 dong (B) 30 000 dong (C) 45 000 dong

(D) 135 000 dong (E) 180 000 dong

Your work:

Problem 8

Which expression produces the values in the second row of the table, given the values of n in the first row?

n	1	2	3	4	5
Value	1	3	5	7	9

(A) $3n - 2$ (B) $2(n - 1)$ (C) $n + 4$ (D) $2n$ (E) $2n - 1$

Your work:

Problem 9

When two numbers are added, the result is -26. One of the numbers is 11. The other number is

(A) -37 (B) 37 (C) -15 (D) 15 (E) -48

Your work:

Problem 10

Mr Hùng has 10 000 000 dong, all in 50 000 dong notes. How many notes does he have?

(A) 1 000 (B) 200 (C) 1 250 (D) 500 (E) 2 000

Your work:

Problem 11

The product $60 \times 60 \times 24 \times 7$ equals

(A) the number of minutes in seven weeks

(B) the number of hours in sixty days

(C) the number of seconds in seven hours

(D) the number of seconds in one week

(E) the number of minutes in twenty-four weeks

Your work:

Problem 12 *

Each of a, b, c, d is a positive integer greater than 3, and

$$\frac{1}{a-2} = \frac{1}{b+2} = \frac{1}{c+1} = \frac{1}{d-3}.$$

Which ordering of these four numbers is correct?

(A) $a < b < c < d$ (B) $c < b < a < d$ (C) $b < a < c < d$

(D) $d < a < c < b$ (E) $b < c < a < d$

Your work:

Problem 13

A rectangle has length x and width y. A triangle has base 16 and corresponding height x. If the area of the rectangle equals the area of the triangle, then the value of y is

(A) 16 (B) 4 (C) 8 (D) 12 (E) 32

Your work:

Problem 14

The symbol $\begin{array}{|c|c|}\hline 3 & 4 \\ \hline 5 & 6 \\ \hline \end{array}$ is evaluated as $3 \times 6 + 4 \times 5 = 38$. If $\begin{array}{|c|c|}\hline 2 & 6 \\ \hline 1 & \\ \hline \end{array}$ is evaluated as 16, then the number that should be placed in the empty space is

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Your work:

Problem 15

Three pumpkins are weighed two at a time in all possible ways. The weights of the pairs are 12 kg, 13 kg and 15 kg. How much does the lightest pumpkin weigh?

(A) 4 kg (B) 5 kg (C) 6 kg (D) 7 kg (E) 8 kg

Your work:

Problem 16

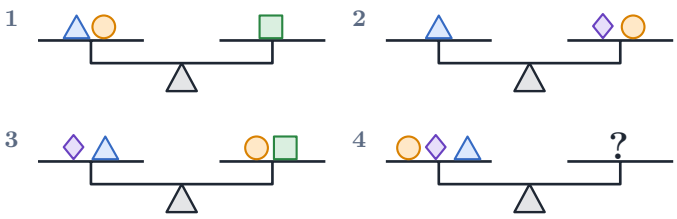
Hồng is learning to save money. Today she has 120 thousand dong in her savings account. From now on, every time she makes a deposit she will add exactly 30 thousand dong. If Hồng makes m such deposits, which expression correctly gives the amount of money (in thousands of dong) in her account?

(A) $120 + m$ (B) $30m$ (C) $30 + 120m$ (D) $150m$ (E) $120 + 30m$

Your work:

Problem 17

All four scales below are balanced. Identical shapes have equal masses. One possible replacement for the "?" that keeps the fourth scale balanced is



(A) △□ (B) ◇△ (C) ○□ (D) □◇ (E) △○

Your work:

Problem 18

The number that goes in the box to make $\frac{3}{7} = \frac{\square}{63}$ true is

(A) 27 (B) 9 (C) 59 (D) 63 (E) 3

Your work:

Problem 19

The sum of four numbers is T. Each of the four numbers is increased by 1. The four new numbers are added together, and the sum is then multiplied by 3. The final result is

(A) $3T + 3$ (B) $3T + 4$ (C) $3T + 12$ (D) $T + 12$ (E) $12T$

Your work:

Problem 20

Two isosceles triangles each have at least one angle measuring $70°$. In the first triangle, the measure (in degrees) of each of the other two angles is even. In the second triangle, the measure of each of the other two angles is odd. Let S be the sum of the two equal angles of the first triangle and T the sum of the two equal angles of the second triangle. The value of $S + T$ is

(A) $280°$ (B) $250°$ (C) $220°$ (D) $200°$ (E) $300°$

Your work:

Problem 21

Ánh is on a 6-day kayak trip with a total distance of 168 km. At the end of the third day she has covered $\frac{3}{7}$ of the total distance. How many kilometres per day must she average for the rest of the trip?

(A) 29 (B) 24 (C) 27 (D) 32 (E) 26

Your work:

Problem 22

A dot starts at the point $(20, 19)$ of the coordinate plane. On the first move it travels 1 unit horizontally or vertically, that is, to one of the points $(21, 19)$, $(19, 19)$, $(20, 20)$ or $(20, 18)$. On the second move it travels 2 units (in either direction) *perpendicular* to the first move. From then on, each move is 1 unit longer than the previous one (3 units, 4 units, 5 units, ...) and is always perpendicular to the move just before it. The dot stops after 10 moves. Which of the following final positions is **not** possible?

(A) $(27, 33)$ (B) $(30, 40)$ (C) $(21, 21)$ (D) $(42, 44)$ (E) $(37, 37)$

Your work:

Problem 23

Giang works as a server in a restaurant. On Saturday she gets up at 6:30 a.m., starts work at x a.m. and finishes at x p.m. (the same day). How long does Giang work on Saturday?

(A) $24 - 2x$ hours (B) $12 - x$ hours (C) $2x$ hours
(D) 0 hours (E) 12 hours

Your work:

Problem 24 ★

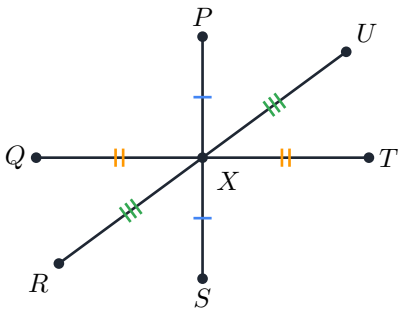
An $8 \times 8 \times n$ rectangular box (n a positive integer) is built from $1 \times 1 \times 1$ cubes. Let A be the surface area of the box and B the combined surface area of all the small cubes that make it up. The sum of all values of n for which $\frac{B}{A}$ is an integer is

(A) 86 (B) 90 (C) 70 (D) 78 (E) 96

Your work:

Problem 25

Three bus routes cross at the stop X as shown (segments with the same number of tick marks have equal length). Every 12 minutes, bus A completes a trip $P \to X \to S \to X \to P$. Every 20 minutes, bus B completes a trip $Q \to X \to T \to X \to Q$. Every 28 minutes, bus C completes a trip $R \to X \to U \to X \to R$. At 1:00 p.m., buses A, B and C leave P, Q and R respectively; each bus drives at a constant speed and turns around instantly at the end of its route. The buses run continuously until 11:00 p.m. Between 5:00 p.m. and 10:00 p.m., how many times are two or more buses at X at the same moment?



(A) 18 (B) 19 (C) 20

(D) 21 (E) 22

Your work:

Problem 26

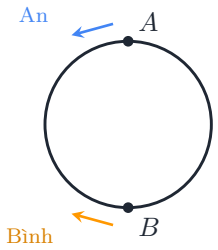
Sơn starts at 7 and counts forward by 3, getting the list $7, 10, 13, \ldots$ Dũng starts at 2026 and counts backward by 5, getting the list $2026, 2021, 2016, \ldots$ Which of the following numbers appears in both lists?

(A) 1009 (B) 1006 (C) 1003 (D) 1001 (E) 1011

Your work:

Problem 27

On a circular track, An stands at point A and Bình stands at point B directly opposite A (segment AB is a diameter). An runs counterclockwise and Bình runs clockwise. They run at constant but different speeds. After a while they notice that every time they pass each other, it is always at the same three places on the track. The ratio of their speeds is



(A) 3 : 2 (B) 3 : 1 (C) 4 : 1

(D) 2 : 1 (E) 5 : 2

Your work:

Problem 28

The list $11, 20, 31, 51, 82$ is an example of an increasing list of five positive integers in which the first and second numbers add to the third, the second and third add to the fourth, and the third and fourth add to the fifth. How many such lists have 124 as the fifth number?

(A) 10 (B) 7 (C) 9 (D) 6 (E) 8

Your work:

Problem 29

Khánh owns more cats than Lan and more dogs than Bảo. Lan owns more dogs than Khánh but fewer cats than Bảo. Which of the following statements **must** be true?

(A) Bảo owns the fewest cats.

(B) Bảo owns the most cats.

(C) Khánh owns the most cats.

(D) Lan owns the most dogs.

(E) Khánh owns the fewest dogs.

Your work:

Problem 30

Linh bought 300 g of candy A for 50 thousand dong and x g of candy B for 70 thousand dong. She worked out that the average price of all the candy she bought was 15 thousand dong per 100 g. What is the value of x?

(A) 525 (B) 600 (C) 500 (D) 450 (E) 900

Your work:

Extension and challenge

Problem 31

The road from Mr Ba's home to the city is 180 km long. He rides a motorbike at a constant speed, and after 3 hours he has covered $\frac{2}{3}$ of the distance.

a) Find the speed of the motorbike.

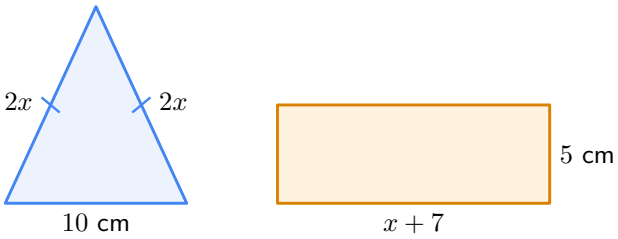
b) If he keeps the same speed, how much longer will it take him to reach the city?

c) If he wants to arrive 30 minutes earlier than planned in part b), at what speed must he ride for the rest of the way?

Your work:

Problem 32

An isosceles triangle has a base of 10 cm and two equal sides of $2x$ cm each. A rectangle has length $(x + 7)$ cm and width 5 cm. The perimeter of the triangle equals the perimeter of the rectangle.



a) Write an expression in x for the perimeter of each shape.

b) Find x, then find the side lengths of the triangle and the area of the rectangle.

Your work:

Problem 33

We use the rule of Problem 14 again: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = a \cdot d + b \cdot c$.

a) Evaluate $\begin{vmatrix} -3 & 4 \\ 5 & 2 \end{vmatrix}$.

b) Find x if $\begin{vmatrix} x & 2 \\ 3 & 4 \end{vmatrix} = \begin{vmatrix} 1 & x \\ 5 & 2 \end{vmatrix}$.

c) Find all integers x such that $\begin{vmatrix} x & x \\ x & x \end{vmatrix} = 50$.

Your work:

Problem 34

a) The sum of five consecutive odd numbers is 155. Find the five numbers.

b) Nam says: "I have found four consecutive odd numbers whose sum is 150." Is Nam right or wrong? Why?

Your work:

Problem 35

Every 2 scoops of fish food are enough to feed 12 fish for one day.

a) How many fish can 5 scoops of food feed for one day?

b) Hà's fish tank has 18 fish. A box of food contains 60 scoops. For how many days will the box feed the whole tank?

Your work:

Problem 36

Each bus ride costs Lan 7 000 dong. The bus company also sells a monthly pass for 200 000 dong with unlimited rides.

a) Write an expression for the amount Lan pays if she makes n rides in a month without a monthly pass.

b) What is the smallest number of rides in a month for which the monthly pass is cheaper than paying per ride?

Your work:

Problem 37

Look at the table below (the values increase steadily as n increases):

n	1	2	3	4	5
Value	2	7	12	17	22

a) Write an expression in n for the row of values.

b) Find the value for $n = 40$.

c) For which n is the value 302?

d) Is any value in the table (continued forever) equal to 2026?

Your work:

Problem 38

a) The product $60 \times 24 \times 365$ is the number of which time unit in a non-leap year (with 365 days)? Work out the product.

b) How many days, hours, minutes and seconds are there in one million seconds?

Your work:

Problem 39

Four positive integers a, b, c, d satisfy

$$\frac{3}{a-5} = \frac{3}{b+1} = \frac{3}{c-2} = \frac{3}{d+4} \qquad \text{and} \qquad a+b+c+d = 50.$$

a) Arrange a, b, c, d in increasing order.

b) Find the value of each number.

Your work:

Problem 40 ⋆⋆

There are four bags of rice with different weights. They are weighed two at a time in all possible ways, giving six results:

7 kg, 8 kg, 9 kg, 10 kg, 11 kg, 12 kg.

a) Find the total weight of the four bags.

b) Each bag weighs a whole number of kilograms. Find the weight of each bag.

Your work:

Problem 41 ⋆

The three scales below are balanced; identical shapes have equal masses.



Scale 1 Scale 2 Scale 3

a) How many □ shapes balance one ○?

b) How many □ shapes balance one ◇?

c) One ◇ and one ○ are placed on the left pan. If only △ shapes may be used on the right pan, how many △ are needed?

d) The four objects △, ○, □, ◇ (one of each kind) have a total mass of 50 g. Find the mass of each object.

Your work:

Problem 42 *

In this problem we only consider triangles in which every angle measures (in degrees) a *whole number*. Two isosceles triangles are said to be of the *same type* if they have the same three angle measures.

a) Prove that there is no triangle whose three angles all have odd measures.

b) How many types of isosceles triangle have all three angles with even measures?

c) How many types of isosceles triangle have exactly two angles with odd measures? Is there an isosceles triangle with exactly one angle of odd measure?

d) How many types of isosceles triangle are there altogether?

Your work:

Problem 43 ★★

A dot starts at the origin $O(0,0)$ and moves by the rule of Problem 22: the first move is 1 unit horizontally or vertically; each later move is 1 unit longer than the previous one and perpendicular to the move just before it (either direction allowed).

a) Explain why after exactly 10 moves the dot cannot be back at O.

b) Find the smallest n ($n \geq 1$) such that after exactly n moves the dot can be back at O. Show one way to do it.

c) Can the dot be back at O after exactly 8 moves?

Your work:

Problem 44 ★★

A $4 \times 6 \times n$ rectangular box (n a positive integer) is built from $1 \times 1 \times 1$ unit cubes. Let A be the surface area of the box and B the combined surface area of all the unit cubes.

a) Write A, B and the ratio $\dfrac{B}{A}$ as expressions in n.

b) Prove that $\dfrac{B}{A} < 7.2$ for every n.

c) Find all values of n for which $\dfrac{B}{A}$ is an integer.

Your work:

Problem 45 **

Two bus routes cross at the stop X. Bus A runs back and forth along the route P–X–S; each arm (PX or XS) takes exactly 4 minutes. Bus B runs along the route Q–X–T; each arm takes exactly 6 minutes. At 1:00 p.m., bus A leaves P and bus B leaves Q; the buses turn around instantly at the ends of their routes and run continuously.

a) Prove that buses A and B are never at X at the same time.

b) A bus C is added on the route R–X–U; each arm takes 10 minutes, and it leaves R at 1:00 p.m. Between 1:00 p.m. and 7:00 p.m., at which times are two buses at X together?

c) Suppose each arm of bus C's route takes m minutes (m any positive integer). Prove that bus C cannot both meet bus A at X (at some moment) and meet bus B at X (at some moment).

Your work:

Problem 46 ⋆

Sơn counts forward $7, 10, 13, \ldots$ (adding 3 each time) and stops before going past 2026. Dũng counts backward $2026, 2021, 2016, \ldots$ (subtracting 5 each time) and stops before going below 7.

a) How many numbers appear in both lists?

b) Find the sum of all the numbers in part a.

c) Mai writes the list of positive multiples of 4: $4, 8, 12, \ldots$ How many numbers appear in all three lists (Sơn's, Dũng's and Mai's)? What is the largest of them?

Your work:

Problem 47 ⋆⋆

An and Bình both start at point A on a circular track; each runs at a constant speed.

a) They run in opposite directions, with the ratio of speeds An : Bình equal to $5 : 3$. At how many different points of the track do they meet?

b) If they run in the *same direction* (still with ratio $5 : 3$), at how many different points does An overtake Bình?

c) Find all speed ratios $p : q$ (with $p > q$, where p and q are coprime positive integers) such that, running in opposite directions, they meet at exactly 10 different points.

Your work:

Problem 48 **

We call an *increasing* list of positive integers a *Fibo list* if, from the third number on, each number is the sum of the two numbers just before it (for example $11, 20, 31, 51, 82$ as in Problem 28).

a) Find all Fibo lists of 6 numbers whose last number is 100.

b) How many Fibo lists of 5 numbers have last number 200?

c) Prove that there is no Fibo list of 6 numbers whose last number is 30.

Your work:

Problem 49 *

Four friends An, Bình, Chi and Dũng each have at least one marble. We know that:

(1) An and Bình together have as many marbles as Chi and Dũng together.

(2) An and Dũng together have more marbles than Bình and Chi together.

(3) Bình has more marbles than An and Chi together.

a) Put the four friends in order from fewest to most marbles.

b) Altogether the four friends have 30 marbles. What is the greatest number of marbles Chi can have? In that case, how many marbles does each friend have?

Your work:

Problem 50

A shop sells candy A at 12 thousand dong per 100 g and candy B at 20 thousand dong per 100 g.

a) How many grams of each kind must be mixed to get 1 kg of mixture costing 15 thousand dong per 100 g (the mixture is worth exactly the total value of the candy put in)?

b) How many grams of candy C, costing 30 thousand dong per 100 g, must be added to the 1 kg mixture of part a to get a new mixture costing 18 thousand dong per 100 g?

c) Prove that if candy A and candy B (both present) are mixed in any proportion, the price of the mixture is always more than 12 and less than 20 thousand dong per 100 g.

Your work: